

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Pb 210, Bi-210, and Po-210			FORMS: Soluble		
PHYSICAL CHARACTERISTICS					
HALF-LIFE: 22.2 years (Pb,) 5 days (Bi), 138.4 days (Po)					
DECAY EMISSIONS					
Assuming secular equilibrium					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
11	11	389	100	5304	100
13	10	30*	55		
47	4	16	16		
		43*	15		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY					
C – level (low hazard): ≤ 2 µCi					
B – level (moderate hazard): > 2 µCi, ≤ 100 µCi					
A – level (high hazard): > 100 µCi					
EXTERNAL RADIATION HAZARDS			INTERNAL RADIATION HAZARDS		
Does not include daughter emissions			Annual Limit on Intake: 0.6 µCi (Ingestion)		
Gamma dose rate, point source at 1 ft, 1 mCi:			0.2 µCi (Inhalation)		
~ 0.04 mrem/h					
Beta dose rate to skin, point source at 1 ft, 1 mCi:			*Highly radiotoxic metal*		
444 mrem/h			The values above indicate the activity taken into the		
Contamination skin dose, uniform deposit of 1 µCi per cm²:			body that would result in either 5 rem to the whole		
6320 mrem/h			body (CEDE) or 50 rem to an organ or tissue (CDE).		
SHIELDING			DOSIMETRY AND BIOASSAY REQs		
Does not include daughter emissions			Whole-body and finger-ring dosimeters are required		
Gammas/X-rays:			for handling 5 mCi or more, or 1 mCi amounts		
<0.1 mm of lead will reduce the gamma dose rate by			weekly. Urine assays may be required after spills or		
90%.			contaminations involving 0.1 µCi.		
Betas/electrons:					
4 mm of plastic will absorb all emissions.					
Bremsstrahlung may be created and require further					
shielding.					
SPECIAL PROBLEMS AND PRECAUTIONS:					
1. Recommended survey probe: Thin-window NaI (low efficiency)					
2. Pb-210 and daughters are highly radiotoxic. Work in a fume hood to capture any dusts.					
3. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often.					
4. Survey work areas before, during, and after work. Instrument and smear surveys are required.					
5. Segregate waste to those with half-lives greater than 120 days (excluding H3 and C14).					
6. Solidify liquid wastes in cement kits.					

References:

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- Johnson, T.E., Birkly, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
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